

JAVA FULL STACK

ENGINEER ROADMAP



A structured path from web fundamentals to enterprise Java

Design -> Web -> Frontend -> Core Java -> Spring -> Full Stack -> Testing

Every project follows the professional Git and code-review workflow.

Roadmap Contents

A focused Java Full Stack learning path derived from the provided Full Stack Engineer roadmap.

Core stack

Frontend: HTML5, CSS3, JavaScript, TypeScript, React Backend: Java, J2EE, Spring, Spring MVC, Spring Boot, Spring Data JPA, Spring Security Data: SQL, PostgreSQL, Hibernate/JPA Quality: Testing

Modules

01. 00 - Engineering Workflow & Git Collaboration
02. 01 - Design Thinking & Figma
03. 02 - Internet & Web Fundamentals
04. 03 - HTML5
05. 04 - CSS3
06. 05 - JavaScript
07. 06 - TypeScript
08. 07 - React
09. 08 - Core Java
10. 09 - Data Structures & Algorithms
11. 10 - Advanced Java (J2EE)
12. 11 - SQL & Relational Databases
13. 12 - Hibernate & JPA
14. 13 - Spring Framework
15. 14 - Spring MVC
16. 15 - Spring Boot
17. 16 - Spring Data JPA
18. 17 - Spring Security & JWT
19. 18 - Full Stack Integration
20. 19 - Testing

Project rule: Every module project must follow the professional repository, branch, commit, review, merge, and versioning workflow described in Module 00.

Engineering Workflow & Git Collaboration

Repository Setup

Step	Command / Naming	Purpose
Fork	GitHub UI -> Fork	Create personal copy
Clone	git clone git@github.com:<you>/fullstack-roadmap.git	Copy to local
Add upstream	git remote add upstream git@github.com:org/fullstack-roadmap.git	Track original
Verify	git remote -v	Confirm remotes
Sync main	git fetch upstream && git checkout main && git merge upstream/main	Stay updated

Branch Strategy

Step	Command / Naming	Purpose
main	main	Production-ready, protected
develop	develop	Integration branch
Feature	feature/JIRA-123-user-auth	New features
Bugfix	bugfix/JIRA-456-login-error	Non-critical fixes
Hotfix	hotfix/JIRA-789-prod-down	Critical production
Release	release/v1.2.0	Release preparation
Chore	chore/update-deps	Maintenance
Docs	docs/api-readme	Documentation
Refactor	refactor/service-layer	Code cleanup
Design	design/figma-login-flow	Design assets

Commit Conventions

Step	Command / Naming	Purpose
feat	feat(auth): add JWT refresh token	New feature
fix	fix(cart): resolve null pointer	Bug fix
docs	docs(readme): update setup steps	Documentation
style	style: apply prettier	Formatting
refactor	refactor(user): extract mapper	Code refactor
perf	perf(query): add index hint	Performance
test	test(auth): add login tests	Tests
build	build: bump spring boot to 3.3	Build system
ci	ci: add sonar step	CI config
chore	chore: update .gitignore	Maintenance

Step	Command / Naming	Purpose
revert	revert: feat(auth) ...	Revert commit
design	design: add login wireframes	Design assets

Code Review & Approval

Step	Command / Naming	Purpose
1 - Author	Open PR	Assign reviewers
2 - CI	Build, tests, lint, security scan	Validate change
3 - Reviewer	Review code and request changes	Quality gate
4 - Designer	Review UI/UX when applicable	Design gate
5 - Author	Address feedback and push fixes	Resolve comments
6 - Reviewer	Approve	Approval
7 - Maintainer	Merge	Integrate change

Review Rules

Step	Command / Naming	Purpose
Require PR	main: Yes develop: Yes	All changes go through PRs
Required approvals	main: 2 develop: 1	Peer review
Design approval	main: Yes (UI) develop: Yes (UI)	UI/UX quality
Dismiss stale approvals	main: Yes develop: Yes	Re-review after updates
Status checks	main: Yes develop: Yes	CI gate
Conversation resolution	main: Yes develop: Yes	Resolve review threads
Signed commits	main: Yes develop: No	Commit authenticity
Linear history	main: Yes develop: Yes	Clean history

Merge Strategies

Step	Command / Naming	Purpose
Squash & Merge	Feature branches	gh pr merge --squash
Rebase & Merge	Linear history preferred	gh pr merge --rebase
Merge Commit	Preserve history	gh pr merge --merge

Release & Versioning

Step	Command / Naming	Purpose
MAJOR	Breaking change	1.0.0 -> 2.0.0
MINOR	New feature	1.0.0 -> 1.1.0

Step	Command / Naming	Purpose
PATCH	Bug fix	1.0.0 -> 1.0.1

Project: Applies to every module project

01 - Design Thinking & Figma

Sub-Module

Sub-Module	Topics
01-01 Design Thinking	Empathize, Define, Ideate, Prototype, Test, Double Diamond, Design Sprint
01-02 UX Research	User Interviews, Surveys, Personas, Empathy Maps, Journey Maps, User Stories
01-03 Information Architecture	Sitemaps, User Flows, Card Sorting, Navigation Patterns
01-04 Wireframing	Low-Fidelity, Mid-Fidelity, High-Fidelity, Sketching, Crazy 8s
01-05 Figma Fundamentals	Frames, Auto Layout, Constraints, Components, Variants, Styles
01-06 Figma Advanced	Design Tokens, Component Sets, Variables, Modes, Prototyping, Smart Animate
01-07 Design Systems	Atomic Design, Tokens, Component Library, Documentation, Storybook
01-08 Prototyping	Clickable Prototypes, Micro-interactions, Transitions, User Testing
01-09 Handoff	Dev Mode, Inspect, Zeplin, Specs, Redlines, Asset Export
01-10 Accessibility Design	Color Contrast (WCAG), Focus States, Touch Targets, Inclusive Design
01-11 Visual Design	Typography, Color Theory, Spacing, Grid Systems, Hierarchy
01-12 Design Critique	Heuristics (Nielsen), Feedback, Iteration, A/B Testing

Project: End-to-End Design of a SaaS Dashboard (Research -> Wireframe -> Hi-Fi -> Prototype -> Handoff)

02 - Internet & Web Fundamentals

Sub-Module

Sub-Module	Topics
02-01 How Internet Works	Client-Server, DNS, HTTP/HTTPS, HTTP/2, HTTP/3 (QUIC), TCP/IP, Request-Response, Status Codes, CDN basics
02-02 Web Browsers	Browser Architecture, Rendering Engine, V8, Browser Storage, DevTools, Lighthouse, Core Web Vitals
02-03 Web Security	HTTPS/TLS, CORS, CSP, XSS, CSRF, SQL Injection, Same-Origin, OWASP Top 10
02-04 Web Protocols	REST, GraphQL, gRPC, WebSocket, SSE, WebRTC
02-05 Version Control	Git Internals, Branching, Merge vs Rebase, Cherry-pick, Bisect, Conventional Commits
02-06 Domain & Hosting	Domain Registrar, DNS Records, SSL/TLS Certificates, CDN, Reverse Proxy

Project: Personal Portfolio with Git Workflow (Fork -> PR -> Review -> Merge)

03 - HTML5

Sub-Module

Sub-Module	Topics
03-01 Fundamentals	Document Structure, Semantic Elements, Metadata, SEO, Open Graph
03-02 Forms	Input Types, Validation, ARIA, File Uploads, FormData API
03-03 Multimedia	Audio, Video, Canvas, SVG, iframe, Picture, Source
03-04 Modern APIs	Web Components, Shadow DOM, Templates, Custom Elements, Web Storage, Drag & Drop
03-05 Accessibility	WCAG 2.2, ARIA Roles, Keyboard Nav, Screen Readers, Focus Management
03-06 PWA Basics	Manifest, Service Workers, Offline, Install Prompt

Project: Accessible Multi-Page Website + PWA

04 - CSS3

Sub-Module

Sub-Module	Topics
04-01 Fundamentals	Selectors, Specificity, Box Model, Units, Variables, Cascade, Inheritance
04-02 Layout	Flexbox, Grid, Positioning, Multi-column, Container Queries, Subgrid
04-03 Responsive	Mobile-First, Media Queries, Fluid Typography, Clamp, Container Units
04-04 Modern CSS	Nesting, :has(), :is(), :where(), Layers, Logical Properties, aspect-ratio, accent-color
04-05 Architecture	BEM, Tailwind, CSS Modules, CSS-in-JS, Atomic CSS
04-06 Animation	Transitions, Keyframes, Transform 3D, Scroll-Driven, View Transitions
04-07 Advanced	CSS Houdini, Paint API, Scroll Snap, Custom Properties, color-mix()

Project: Responsive Admin Dashboard with Dark Mode + Animations

05 - JavaScript

Sub-Module

Sub-Module	Topics
05-01 Fundamentals	Variables, Data Types, Coercion, Operators, Control Flow, Strict Mode
05-02 Functions	Declarations, Arrow, Closures, IIFE, HOF, Currying, Memoization
05-03 Objects	Creation Patterns, Prototype Chain, Classes, Getters/Setters, Symbol, WeakMap/WeakSet
05-04 Arrays	map/filter/reduce, Destructuring, Spread/Rest, Iterators, Generators
05-05 Modern JS	Template Literals, Modules, Optional Chaining, Nullish Coalescing, Top-Level Await
05-06 Async	Event Loop, Callbacks, Promises, async/await, Combinators, AbortController
05-07 DOM	Selection, Traversal, Events, Delegation, Forms, MutationObserver
05-08 Browser APIs	Fetch, Storage, IndexedDB, Web Workers, IntersectionObserver, ResizeObserver, Geolocation, Clipboard
05-09 Meta Programming	Proxy, Reflect, Symbol, Decorators
05-10 Error Handling	try/catch, Custom Errors, Global Handlers, Error Boundaries
05-11 Memory	Garbage Collection, Memory Leaks, Heap Snapshots

Project: Task Management SPA (Vanilla JS)

06 - TypeScript

Sub-Module

Sub-Module	Topics
06-01 Fundamentals	Type System, Inference, Interfaces vs Types, Enums, Tuples, any vs unknown
06-02 Advanced Types	Union, Intersection, Generics, Conditional, Mapped, Template Literal, Discriminated Unions
06-03 Utility Types	Partial, Required, Pick, Omit, Record, Readonly, Exclude, Extract, ReturnType
06-04 TypeScript + React	Props, Hooks, Events, Context, Generic Components
06-05 Tooling	tsconfig, Strict Mode, Declaration Files, Type Guards, satisfies, const assertions
06-06 Advanced Patterns	Branded Types, Builder Types, Type-Level Programming

Project: Type-Safe API Client Library

07 - React

Sub-Module

Sub-Module	Topics
07-01 Fundamentals	Architecture, JSX, Components, Props, State, Lifecycle
07-02 Hooks	useState, useEffect, useContext, useReducer, useRef, useMemo, useCallback, useLayoutEffect, useId, useTransition, useDeferredValue, useSyncExternalStore, Custom Hooks
07-03 Patterns	Compound, Render Props, HOC, Controlled/Uncontrolled, Composition, Slots
07-04 State Management	Local, Context, Redux Toolkit, RTK Query, Zustand, TanStack Query, Jotai
07-05 Routing	React Router v7, Nested, Dynamic, Protected, Data Loaders, Lazy
07-06 Forms	Controlled, React Hook Form, Zod, File Uploads, Multi-step
07-07 API	Fetch, Axios, Error Handling, Cancellation, Optimistic Updates, Infinite Queries
07-08 Performance	React.memo, useMemo/useCallback, Code Splitting, Virtualization, Profiler, Concurrent Features
07-09 Testing	Jest, RTL, Hooks Testing, MSW, Integration Tests
07-10 Production	Error Boundaries, Suspense, RSC, Portals, Build Optimization
07-11 Advanced	Server Components, Streaming, Suspense Boundaries, Transitions
07-12 Animation	Framer Motion, React Spring, Layout Animations

Project: Enterprise React Admin Portal (JWT, RBAC, Charts, Realtime)

08 - Core Java

Sub-Module

Sub-Module	Topics
08-01 Fundamentals	JDK/JRE/JVM, Variables, Types, Operators, Control Flow, Arrays, var
08-02 OOP	Classes, Objects, Encapsulation, Inheritance, Polymorphism, Abstraction, Interfaces
08-03 Advanced OOP	Inner Classes, Static, Final, Packages, Access Modifiers, Records, Sealed Classes, Pattern Matching
08-04 Exceptions	Hierarchy, Checked/Unchecked, Custom, try-with-resources, Multi-catch
08-05 Generics	Classes, Methods, Bounded Types, Wildcards, Type Erasure
08-06 Collections	List, Set, Map, Queue, Deque, Comparable/Comparator, Concurrent Collections
08-07 Functional	Lambda, Functional Interfaces, Method Refs, Streams, Optional
08-08 Concurrency	Threads, Runnable, Callable, Synchronization, Locks, ExecutorService, CompletableFuture, Atomic, Virtual Threads, Structured Concurrency
08-09 I/O & NIO	File I/O, BufferedReader/Writer, Serialization, NIO Channels, Path/Files
08-10 JVM	Memory Model, GC (G1, ZGC), JIT, JVM Tuning, JMX
08-11 Modules (JPMS)	module-info, Requires, Exports
08-12 New Language Features	Text Blocks, Switch Expressions, Records, Sealed Classes

Project: Banking System with Concurrency

09 - Data Structures & Algorithms

Sub-Module

Sub-Module	Topics
09-01 Complexity	Big O, Time/Space, Amortized
09-02 Arrays & Strings	Two Pointers, Sliding Window, Prefix Sum, Kadane, String Manipulation
09-03 Linked Lists	Singly, Doubly, Circular, Fast/Slow, Reversal
09-04 Stacks & Queues	Stack, Queue, Deque, Monotonic Stack, Priority Queue
09-05 Hashing	Hash Tables, Collision, HashMap/HashSet, Frequency
09-06 Trees	Binary, BST, Traversals, AVL, Red-Black, Segment, Fenwick, Trie
09-07 Heaps	Min/Max, Heapify, Heap Sort, Top-K
09-08 Graphs	Representation, BFS, DFS, Topological, Dijkstra, Bellman-Ford, Floyd-Warshall, MST, Union-Find
09-09 Recursion	Patterns, Subsets, Permutations, Combinations, N-Queens, Sudoku
09-10 DP	Memoization, Tabulation, 1D/2D, Knapsack, LCS, LIS, Matrix Chain, DP on Trees
09-11 Advanced	Bit Manipulation, Greedy, Divide & Conquer, Intervals, Segment Trees, Sqrt Decomposition
09-12 Problem Patterns	Sliding Window, Two Pointers, Fast/Slow, Merge Intervals, Cyclic Sort, Top-K, K-way Merge

Project: 300+ LeetCode Problems

10 - Advanced Java (J2EE)

Sub-Module

Sub-Module	Topics
10-01 JDBC	Architecture, DriverManager, Connection Pooling, PreparedStatement, ResultSet, Batch, Transactions
10-02 Servlets	Lifecycle, Request/Response, Sessions, Cookies, Filters, Listeners, ServletContext
10-03 JSP	Lifecycle, Scriptlets, EL, JSTL, Custom Tags
10-04 MVC	Model, View, Controller, Request Flow

Project: MVC Web Application (Servlet + JSP + JDBC)

11 - SQL & Relational Databases

Sub-Module

Sub-Module	Topics
11-01 Fundamentals	Concepts, Tables, Rows, Columns, Keys, Constraints
11-02 CRUD	SELECT, INSERT, UPDATE, DELETE, UPSERT, MERGE
11-03 Querying	WHERE, ORDER BY, GROUP BY, HAVING, DISTINCT, LIMIT/OFFSET
11-04 Joins	INNER, LEFT/RIGHT, FULL OUTER, CROSS, Self
11-05 Advanced	Subqueries, CTEs, Window Functions, Views, Materialized Views, Stored Procedures, Triggers
11-06 Design	Normalization (1NF-BCNF), ER Diagrams, Relationships, Denormalization
11-07 Transactions	ACID, COMMIT/ROLLBACK, Isolation Levels, Deadlocks
11-08 Performance	Indexes, Query Optimization, EXPLAIN, Partitioning, Connection Pooling
11-09 PostgreSQL	JSON/JSONB, Arrays, Full-Text Search, Extensions, Logical Replication

Project: Enterprise HR Database

12 - Hibernate & JPA

Sub-Module

Sub-Module	Topics
12-01 ORM	Concepts, Architecture, Session/SessionFactory, Entity Lifecycle
12-02 Mapping	@Entity, @Table, @Id, @GeneratedValue, @Column, @Transient, @Embeddable
12-03 Relationships	@OneToOne, @OneToMany, @ManyToOne, @ManyToMany, Cascade, Fetch
12-04 Querying	HQL, JPQL, Criteria API, Native, Named Queries
12-05 Advanced	Caching (1st/2nd), Lazy Loading, Fetch Strategies, Batch, Interceptors, Auditing
12-06 Performance	N+1, Fetch Joins, Entity Graphs, Projections, Batching

Project: Employee Management System

13 - Spring Framework

Sub-Module

Sub-Module	Topics
13-01 Core	IoC, DI, Beans, Scopes, ApplicationContext, Bean Lifecycle
13-02 Configuration	XML, Java Config, Annotations, Component Scanning, Profiles
13-03 AOP	Aspect, Advice, Pointcuts, Join Points, Proxies
13-04 Data Access	Spring JDBC, Transaction Management, @Transactional, ORM Integration
13-05 Events	ApplicationEvent, @EventListener, Async Events
13-06 Scheduling	@Scheduled, TaskScheduler, Quartz

Project: Spring Core Application

14 - Spring MVC

Sub-Module

Sub-Module	Topics
14-01 Architecture	DispatcherServlet, Handler Mapping, View Resolver, Request Flow
14-02 Controllers	@Controller, @RestController, @RequestMapping, @GetMapping, @PostMapping, @PathVariable, @RequestParam, @RequestBody
14-03 Validation	Bean Validation, @Valid, BindingResult, Custom Validators
14-04 Exception Handling	@ExceptionHandler, @ControllerAdvice, ResponseEntity
14-05 REST	Principles, Methods, Status Codes, Content Negotiation, HATEOAS

Project: Spring MVC REST API

15 - Spring Boot

Sub-Module

Sub-Module	Topics
15-01 Fundamentals	Auto-Configuration, Starters, Embedded Server, CLI
15-02 Config	application.properties/yml, Profiles, @ConfigurationProperties, Env Vars
15-03 REST	Controllers, DTOs, Validation, Exception Handling, HATEOAS
15-04 Production	Actuator, Logging (Logback), Health, Metrics, Graceful Shutdown
15-05 Docs	OpenAPI 3, Swagger, SpringDoc
15-06 Testing	@SpringBootTest, @WebMvcTest, @DataJpaTest, MockMvc
15-07 Advanced	DevTools, Native Image (GraalVM), Virtual Threads, AOT

Project: Production-Ready REST API

16 - Spring Data JPA

Sub-Module

Sub-Module	Topics
16-01 Fundamentals	EntityManager, Persistence Context, Entity States
16-02 Repositories	JpaRepository, CrudRepository, PagingAndSorting, Derived Queries, @Query
16-03 Relationships	All Mappings, Cascade, Orphan Removal
16-04 Advanced	Specifications, Query by Example, Projections, Pagination, Sorting
16-05 Performance	Lazy vs Eager, Fetch Joins, N+1, Entity Graphs, Batch
16-06 Auditing	@CreatedDate, @LastModifiedDate, @CreatedBy
16-07 Soft Delete	@SQLDelete, @Where

Project: Enterprise Backend with JPA

17 - Spring Security & JWT

Sub-Module

Sub-Module	Topics
17-01 Fundamentals	AuthN vs AuthZ, Filter Chain, SecurityContext
17-02 Authentication	UserDetailsService, PasswordEncoder, AuthenticationManager, Provider
17-03 JWT	Structure, Generation, Validation, Filter, Refresh Tokens, Stateless
17-04 Authorization	Roles, Authorities, Method Security, @PreAuthorize, RBAC, ABAC
17-05 OAuth2 & OIDC	Flows, Resource Server, Social Login, Keycloak
17-06 Best Practices	Hashing, CORS, CSRF, Headers, Rate Limiting, mTLS

Project: JWT-Secured Enterprise API

18 - Full Stack Integration

Layer

Sub-Module	Topics
Frontend	React + TS, Redux Toolkit, React Router, RHF, Axios Interceptors
Backend	Spring Boot, Security, JPA, JWT, Validation
Integration	REST, Auth Flow, Error Handling, CORS, Pagination, File Upload, Realtime

Project: Full Stack Enterprise Management Portal (Auth, Employee/Client/Project Mgmt, Kanban, Dashboard, RBAC, Realtime)

19 - Testing

Sub-Module

Sub-Module	Topics
19-01 Unit (Java)	JUnit 5, Assertions, Lifecycle, Parameterized, Test Doubles
19-02 Mocking	Mockito, Mocks/Spies/Stubs, Matchers, Verification
19-03 Spring	@SpringBootTest, @WebMvcTest, @DataJpaTest, MockMvc, TestRestTemplate
19-04 Integration	Testcontainers, PostgreSQL, Kafka, MongoDB
19-05 Frontend	Jest, RTL, User Event, MSW
19-06 E2E	Playwright, Cypress, POM
19-07 API	Postman, REST Assured, Newman
19-08 Contract	Pact, Spring Cloud Contract
19-09 Performance	JMeter, Gatling, k6
19-10 Best Practices	AAA, Isolation, Coverage, Mutation Testing (PIT)

Project: Full-stack automated test suite covering Java, Spring, API, frontend, E2E, contract, and performance testing